

Instituto Tecnológico de Costa Rica

Operations Research - Semester II

Simplex

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Simplex

The simplex algorithm was invented by Robert Dantzig in 1947 to solve Linear Programming problems. Dantzig proved that the feasible region (area that satisfies all constraints) of a Linear Programming problem forms a convex set. He also proved that the optimal solution of a LP problem is located at a vertex of the convex body of the feasible region.

Dantzig then used this theorems to create Simplex, an algorithm that moves through the vertices of the feasible convex body until finding the optimal solution to the problem. Simplex is a mechanical method that uses matrix operations to determine the solution.

To setup the simplex algorithm we will:

- For each $\leq$ constraint add a slack variable s_i
- For each $\geq$ constraint add an excess variable e_i and an artificial variable a_i .
- for each $=$ constraint add an artificial variable a_i
- for each artificial variable added:
 - Add Ma_i to the objective function if minimizing.
 - Add $-Ma_i$ to the objective function if maximizing.
 - M is a constant of a large value, but not infinite.

Example

$$\begin{aligned}\text{Maximize } Z &= x_1 + x_2 \\ x_1 + 2x_2 &\leq 3 \\ 4x_1 + 5x_2 &\geq 6 \\ 7x_1 + 8x_2 &= 9\end{aligned}$$

Converts to

$$\begin{aligned}\text{Maximize } Z &= x_1 + x_2 - Ma_1 - Ma_2 \\ x_1 + 2x_2 + s_1 &\leq 3 \\ 4x_1 + 5x_2 + e_1 + a_1 &\geq 6 \\ 7x_1 + 8x_2 + a_2 &= 9\end{aligned}$$

Simplex Table

Using the same example as above, we should construct a matrix called Simplex Table as follows:

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	-1	-1	0	0	M	M	0
	0	1	2	1	0	0	0	3
	0	4	5	0	-1	1	0	6
	0	7	8	0	0	0	1	9

Execution of Simplex algorithm:

1. If there are any artificial variables, our first step should be to remove the Ms in their columns through matrix operations, leaving canonical vectors on artificial variable columns.
2. if there are any negative
textbackslash positive values in the first row (excluding Z and b) when maximizing
textbackslash minimizing (respectively)
 - (a) Identify the pivot
 - i. Indetify a column X with the most negative
textbackslash positive (maximizing
textbackslash minimizing), divide all values on column b with their re-
spective value on X.
 - ii. Exclude 0s and negative values
 - iii. The pivot will be the cell on column X on the row with the **smallest**
division result.
 - (b) Through matrix operations, create a canonical vector on the column of the
pivot, leaving 1 exactly on the pivot cell.
 - (c) Go to 2
3. The simplex table will contain the optimal solution.
 - In column b, the first cell shows the value of Z.
 - Every variable with a non-canonical vector in their column has a value of 0.
 - Variables with a canonical vector in their column have a value of what the b
column has on the row where they have a 1.

BebiCo

Minimize

$$Z = 2.00x_1 + 3.00x_2$$

Subject to

$$0.50x_1 + 0.25x_2 \leq 4.00$$

$$1.00x_1 + 3.00x_2 \geq 20.00$$

$$1.00x_1 + 1.00x_2 = 10.00$$

Simplex Table

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	-2	-3	0	0	-M	-M	0
	0	0.5	0.25	1	0	0	0	4
	0	1	3	0	-1	1	0	20
	0	1	1	0	0	0	1	10

Artificial Variables "Canonization"

Variable a_1

$$R_1 \leftarrow R_1 + 20MR_3$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	1M	3M	0	-M	0	-M	20M
	0	0.5	0.25	1	0	0	0	4
	0	1	3	0	-1	1	0	20
	0	1	1	0	0	0	1	10

Variable a_2

$$R_1 \leftarrow R_1 + 10MR_4$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	2M	4M	0	-M	0	0	30M
	0	0.5	0.25	1	0	0	0	4
	0	1	3	0	-1	1	0	20
	0	1	1	0	0	0	1	10

Intermediate Tables

Pivoting 1

Most Positive

Column 3 (4M)

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	2M	4M	0	-M	0	0	30M
	0	0.5	0.25	1	0	0	0	4
	0	1	3	0	-1	1	0	20
	0	1	1	0	0	0	1	10

Fractions

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b	Frac
	1	2M	4M	0	-M	0	0	30M	
	0	0.5	0.25	1	0	0	0	4	16.00
	0	1	3	0	-1	1	0	20	6.67
	0	1	1	0	0	0	1	10	10.00

$$4.00/0.25 = 16.00$$

$$20.00/3.00 = 6.67$$

$$10.00/1.00 = 10.00$$

Smallest fraction: 6.67 $\rightarrow$ Pivot: row 3

Pivot

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	2M	4M	0	-M	0	0	30M
	0	0.5	0.25	1	0	0	0	4
	0	1	3	0	-1	1	0	20
	0	1	1	0	0	0	1	10

Canonization

$$R_3 \leftarrow R_3/3.00$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	2M	4M	0	-M	0	0	30M
	0	0.5	0.25	1	0	0	0	4
	0	0.333333	1	0	-0.333333	0.333333	0	6.66667
	0	1	1	0	0	0	1	10

$$R_1 \leftarrow R_1 + -3999997.00R_3$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M
	0	0.5	0.25	1	0	0	0	4
	0	0.333333	1	0	-0.333333	0.333333	0	6.66667
	0	1	1	0	0	0	1	10

$$R_2 \leftarrow R_2 + -0.25R_3$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M
	0	0.416667	0	1	0.0833333	-0.0833333	0	2.33333
	0	0.333333	1	0	-0.333333	0.333333	0	6.66667
	0	1	1	0	0	0	1	10

$$R_4 \leftarrow R_4 + -1.00R_3$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M
	0	0.416667	0	1	0.0833333	-0.0833333	0	2.33333
	0	0.333333	1	0	-0.333333	0.333333	0	6.66667
	0	0.666667	0	0	0.333333	-0.333333	1	3.33333

Pivot Result

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M
	0	0.416667	0	1	0.0833333	-0.0833333	0	2.33333
	0	0.333333	1	0	-0.333333	0.333333	0	6.66667
	0	0.666667	0	0	0.333333	-0.333333	1	3.33333

Pivoting 2

Most Positive

Column 2 (0.667M)

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M
	0	0.416667	0	1	0.0833333	-0.0833333	0	2.33333
	0	0.333333	1	0	-0.333333	0.333333	0	6.66667
	0	0.666667	0	0	0.333333	-0.333333	1	3.33333

Fractions

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b	Frac
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M	
	0	0.416667	0	1	0.0833333	-0.0833333	0	2.33333	5.60
	0	0.333333	1	0	-0.333333	0.333333	0	6.66667	20.00
	0	0.666667	0	0	0.333333	-0.333333	1	3.33333	5.00

$$2.33/0.42 = 5.60$$

$$6.67/0.33 = 20.00$$

$$3.33/0.67 = 5.00$$

Smallest fraction: 5.00 $\rightarrow$ Pivot: row 4

Pivot

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M
	0	0.416667	0	1	0.0833333	-0.0833333	0	2.33333
	0	0.333333	1	0	-0.333333	0.333333	0	6.66667
	0	0.666667	0	0	0.333333	-0.333333	1	3.33333

Canonization

$$R_4 \leftarrow R_4 / 0.67$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M
	0	0.416667	0	1	0.0833333	-0.0833333	0	2.33333
	0	0.333333	1	0	-0.333333	0.333333	0	6.66667
	0	1	0	0	0.5	-0.5	1.5	5

$$R_1 \leftarrow R_1 + -666665.67R_4$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0	0	0	-0.5	-1M	-1M	25
	0	0.416667	0	1	0.0833333	-0.0833333	0	2.33333
	0	0.333333	1	0	-0.333333	0.333333	0	6.66667
	0	1	0	0	0.5	-0.5	1.5	5

$$R_2 \leftarrow R_2 + -0.42R_4$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0	0	0	-0.5	-1M	-1M	25
	0	0	0	1	-0.125	0.125	-0.625	0.25
	0	0.333333	1	0	-0.333333	0.333333	0	6.66667
	0	1	0	0	0.5	-0.5	1.5	5

$$R_3 \leftarrow R_3 + -0.33R_4$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0	0	0	-0.5	-1M	-1M	25
	0	0	0	1	-0.125	0.125	-0.625	0.25
	0	0	1	0	-0.5	0.5	-0.5	5
	0	1	0	0	0.5	-0.5	1.5	5

Pivot Result

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0	0	0	-0.5	-1M	-1M	25
	0	0	0	1	-0.125	0.125	-0.625	0.25
	0	0	1	0	-0.5	0.5	-0.5	5
	0	1	0	0	0.5	-0.5	1.5	5

Results

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0	0	0	-0.5	-1M	-1M	25
	0	0	0	1	-0.125	0.125	-0.625	0.25
	0	0	1	0	-0.5	0.5	-0.5	5
	0	1	0	0	0.5	-0.5	1.5	5

Objective Value

$$Z = 25.00$$

Variables

$$x_1 = 5.00$$

$$x_2 = 5.00$$

Slack or Surplus

$$s_1 = 0.25$$

$$s_2 = 0.00$$

$$s_3 = 0.00$$

Excess

$$e_1 = 0.00$$